

Project: WelfareLink - Social Welfare & Benefits Distribution System

1. Introduction

WelfareLink is a web-based platform designed for social welfare departments, municipalities, and government agencies to manage welfare programs, benefits distribution, and compliance. It enables citizens to register, apply for welfare schemes, and track disbursements, while administrators and officers can monitor compliance, manage programs, and generate reports.

The system supports workflows for citizen registration, welfare application submission, benefit allocation, disbursement tracking, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across welfare governance.

Actors / Users

- **Citizen:** Registers, applies for welfare programs, tracks benefits.
- **Welfare Officer:** Validates applications, allocates benefits, updates status.
- **Program Manager:** Oversees welfare programs, monitors budgets, tracks performance.
- **Administrator:** Configures workflows, manages users, oversees reporting.
- **Compliance Officer:** Ensures adherence to welfare policies, audits records.
- **Government Auditor:** Reviews compliance reports, monitors welfare program utilization.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Citizen Registration & Profile Management
- 2.3 Welfare Application & Eligibility Validation
- 2.4 Benefit Allocation & Disbursement Management
- 2.5 Program & Resource Management
- 2.6 Compliance & Audit Management
- 2.7 Reporting & Analytics
- 2.8 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.
- **Backend:** REST API-based microservices (Identity, Citizens, Applications, Benefits, Programs, Compliance, Reports).

- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).
- **Deployment:** Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features

- Secure authentication and role-based access control.
- Audit logging of all actions.

Entities

- **User**(UserID, Name, Role [Citizen/Officer/Manager/Admin/Compliance/Auditor], Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)

4.2 Citizen Registration & Profile Management

Features

- Registers citizens and maintains profiles.
- Validates identity and residence documents.

Entities

- **Citizen**(CitizenID, Name, DOB, Gender, Address, ContactInfo, Status)
- **CitizenDocument**(DocumentID, CitizenID, DocType [IDProof/Residence], FileURI, UploadedDate, VerificationStatus)

4.3 Welfare Application & Eligibility Validation

Features

- Citizens apply for welfare programs.
- Officers validate eligibility and track application status.

Entities

- **WelfareApplication**(ApplicationID, CitizenID, ProgramID, SubmittedDate, Status)
- **EligibilityCheck**(CheckID, ApplicationID, OfficerID, Result, Date, Notes)

4.4 Benefit Allocation & Disbursement Management

Features

- Allocate benefits to citizens.
- Track disbursements and utilization.

- Monitor benefit distribution efficiency.

Entities

- **Benefit**(BenefitID, ApplicationID, Type [Cash/Food/Medical], Amount, Date, Status)
- **Disbursement**(DisbursementID, BenefitID, CitizenID, OfficerID, Date, Status)

4.5 Program & Resource Management

Features

- Manage welfare programs.
- Track resources and utilization.
- Monitor program outcomes.

Entities

- **Program**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type [Funds/Materials], Quantity, Status)

4.6 Compliance & Audit Management

Features

- Ensure adherence to welfare policies.
- Maintain compliance records.
- Conduct audits with findings.

Entities

- **ComplianceRecord**(ComplianceID, EntityID, Type [Application/Benefit/Program], Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)

4.7 Reporting & Analytics

Features

- Generate dashboards for applications, benefits, programs, and compliance.
- Provide analytics on welfare program efficiency.

Entities

- **Report**(ReportID, Scope [Application/Benefit/Program], Metrics, GeneratedDate)

4.8 Notifications & Alerts

Features

- Send alerts for application status, benefit disbursement updates, compliance deadlines.
- Notify citizens about welfare program updates.

Entities

- **Notification**(NotificationID, UserID, EntityID, Message, Category [Application/Benefit/Program/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local**: Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production**: Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables & Attributes

- **User**(UserID, Name, Role, Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)
- **Citizen**(CitizenID, Name, DOB, Gender, Address, ContactInfo, Status)
- **CitizenDocument**(DocumentID, CitizenID, DocType, FileURI, UploadedDate, VerificationStatus)
- **WelfareApplication**(ApplicationID, CitizenID, ProgramID, SubmittedDate, Status)
- **EligibilityCheck**(CheckID, ApplicationID, OfficerID, Result, Date, Notes)
- **Benefit**(BenefitID, ApplicationID, Type, Amount, Date, Status)
- **Disbursement**(DisbursementID, BenefitID, CitizenID, OfficerID, Date, Status)
- **Program**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type, Quantity, Status)
- **ComplianceRecord**(ComplianceID, EntityID, Type, Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)
- **Report**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Citizen Portal**: Welfare application submission, benefit tracking, notifications.

- **Welfare Officer Dashboard:** Application validation, benefit allocation, compliance tracking.
- **Program Manager Console:** Program oversight, resource monitoring, reporting.
- **Administrator Panel:** Workflow setup, compliance monitoring, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.
- **Government Auditor Dashboard:** Welfare program review, compliance monitoring.

8. Non-Functional Requirements

- **Performance:** Handle 200,000 concurrent users across welfare networks.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support nationwide rollout across multiple welfare programs.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (application approval rate, benefit distribution rate, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single welfare program before nationwide expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB.